THE PULLBACK OF A SMALL ETALE SHEAF TO THE H-TOPOLOGY IS DESCRIBED BY THE SECTIONS OF ITS SMALL PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the etale description and the deduction of the h-sheaf property from the ph statement.

Editorial note. The label lemma-describe-pullback-pi-h names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The description of the pullback in the etale topology. Same chapter, the lemma identifying pi_S^{-1}F with the rule (f : X -> S) |-> Gamma(X, f_small^{-1}F) for the etale topology. Supplies the etale part of the statement.
- The description of the pullback in the stronger ph topology. Same chapter, the corresponding lemma for the ph topology. Supplies that the same rule is a sheaf in a topology stronger than h.

Source reference. Original reference 09XN: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4960–4960. Prerequisite pointer invoked at proof line(s) 22569; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DDW: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 22175–22175. Prerequisite pointer invoked at proof line(s) 22571; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation as above. Let $\mathcal{F}$ be a sheaf on $S_{étale}$. The rule*

$$(Sch/S)_h \longrightarrow Sets, \quad (f : X \to S) \longmapsto \Gamma(X, f_{small}^{-1}\mathcal{F})$$

is a sheaf and a fortiori a sheaf on $(Sch/S)_{étale}$. In fact this sheaf is equal to $\pi_S^{-1}\mathcal{F}$ on $(Sch/S)_{étale}$ and $\epsilon_S^{-1}\pi_S^{-1}\mathcal{F}$ on $(Sch/S)_h$.

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Proof. The statement about the étale topology is the content of Lemma 09XN. To finish the proof it suffices to show that $\pi_S^{-1}\mathcal{F}$ is a sheaf for the h topology. However, in Lemma 0DDW we have shown that $\pi_S^{-1}\mathcal{F}$ is a sheaf even in the stronger ph topology. □